

Pattern Matching and Machine Learning for Audio Signal Processing

Prof. Dr. Frank Kurth

INSTITUTE OF COMPUTER SCIENCE, UNIVERSITY OF BONN

Chapter 2: Fourier Transforms: DFT, FFT, and STFT

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Goals of this chapter

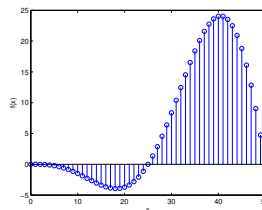
- to introduce and illustrate the Discrete Fourier Transform (DFT)
- to compile fundamental properties of DFTs
- to review the Cooley-Tukey FFT
- to discuss the Short Time Fourier Transform (STFT)
- to present first striking applications

Digital signals: local view

- Suppose, we have a discrete-time signal $\mathbf{f}$ of finite length N .
- Then $\mathbf{f}$ is a sequence of N real or complex numbers:

$$\mathbf{f} = (f(0), f(1), \dots, f(N-1))^{\top}.$$

- Illustration of a real-valued $\mathbf{f}$ as a stem plot:

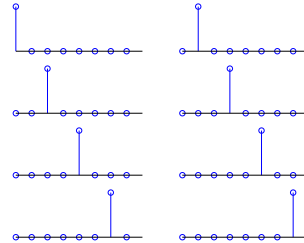


- In general, $\mathbf{f}$ is a linear combination of unit vectors $\mathbf{e}_k$:

$$\boxed{\mathbf{f} = \sum_{k=0}^{N-1} f(k) \mathbf{e}_k}, \quad \text{where} \quad \mathbf{e}_k = \underbrace{(0, \dots, 0, 1, 0, \dots, 0)}_{\substack{k \quad N-k-1}}^{\top} \quad \text{and} \quad f(k) \in \mathbb{C}.$$

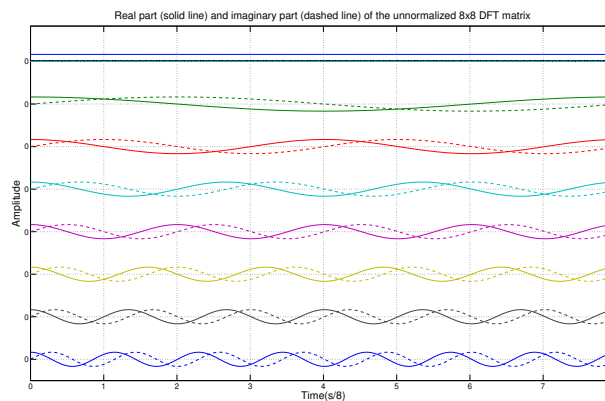
Digital signals: local vs. global view

- The unit vectors $e_0, e_1, \dots, e_{N-1}$ form an orthonormal basis of $\mathbb{C}^N$, however each summand $f(k)e_k$ provides only **extremely local** information about f .
- In numerous applications, another ON-basis of $\mathbb{C}^N$ plays a fundamental role.
- The new basis elements take up an **extremely global** view and are closely related to the **discrete Fourier transform** (DFT).
- The following figures oppose the local and the global basis functions (for $N = 8$).
- Our first figure shows the stem plots of the eight unit vectors in $\mathbb{C}^8$:



Continuous time background of DFT

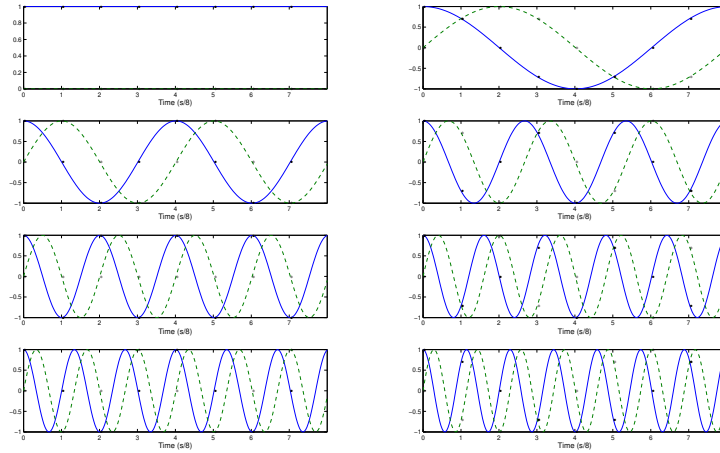
- For $N = 8$, eight pairs of waves (cosine=solid line, sine=dashed line) of 0 Hz, 1 Hz, $\dots$, 7 Hz form the continuous time background of the global view to be described below.



- Sampling these pairs at $0, 1/8, \dots, 7/8$ leads to the 8×8 DFT.

1/8-sampled waves

- Let $k \in [0 : 7]$. Then 1/8-sampling of the cosine (resp. sine) of k Hz gives the real (resp. imaginary) part of the k th basis vector in $\mathbb{C}^8$.



1/N-sampling of exponential signals

- For $k \in [0 : N - 1]$ the k th pair of waves corresponds to the exponential signal

$$[0, 1] \ni t \mapsto \exp(2\pi i k t) = \cos(2\pi k t) + i \cdot \sin(2\pi k t).$$

- For $k, n \in [0 : N - 1]$ define $\mathbf{u}_k \in \mathbb{C}^N$ by

$$\mathbf{u}_k(n) := \exp(2\pi i k n / N).$$

Then $\mathbf{u}_k$ is the 1/N-sampled and truncated version of the k Hz signal $\mathbb{R} \ni t \mapsto \exp(2\pi i k t)$, see the last slide.

- Our next goal is to show that the $\mathbf{u}_k$ are **pairwise orthogonal**, i.e., $\langle \mathbf{u}_k | \mathbf{u}_\ell \rangle = 0$ for $k \neq \ell$, where

$$\langle \mathbf{x} | \mathbf{y} \rangle := \sum_{n=0}^{N-1} x(n) \overline{y(n)} = \mathbf{x}^\top \cdot \overline{\mathbf{y}}$$

denotes the **inner product** of $\mathbf{x}, \mathbf{y} \in \mathbb{C}^N$. Moreover, all $\mathbf{u}_k$ are of Euclidean length $\sqrt{N}$. Thus after normalization, we obtain an orthonormal basis of $\mathbb{C}^N$, the **Fourier basis**.

The Fourier basis

Theorem 1. *The set*

$$\{\mathbf{u}_k / \sqrt{N} \mid k \in [0 : N - 1]\}$$

*forms an orthonormal basis (the **Fourier basis**) of $\mathbb{C}^N$.*

Proof. It suffices to show that $\langle \mathbf{u}_k | \mathbf{u}_\ell \rangle = N \cdot \delta_{k,\ell}$, for $k, \ell \in [0 : N - 1]$.

Using the functional equation $\exp(z + w) = \exp(z) \cdot \exp(w)$, we obtain

$$\exp(2\pi i k n / N) = \exp(2\pi i / N)^{k \cdot n} =: \rho^{k \cdot n},$$

where $\rho := \exp(2\pi i / N)$ is a **primitive N -th root of unity**, i.e., $\rho^m = 1$ iff $m \equiv 0 \pmod{N}$. Thus with

$\sum_{n=0}^{N-1} q^n = (1 - q^N) / (1 - q)$, valid for any $1 \neq q \in \mathbb{C}$, we get for $k \neq \ell$ at first $\rho^{k-\ell} \neq 1$ and hence

$$\langle \mathbf{u}_k | \mathbf{u}_\ell \rangle = \sum_{n=0}^{N-1} \rho^{(k-\ell)n} = \sum_{n=0}^{N-1} (\rho^{k-\ell})^n = \frac{1 - (\rho^{k-\ell})^N}{1 - \rho^{k-\ell}} = \frac{1 - (\rho^N)^{k-\ell}}{1 - \rho^{k-\ell}} = 0.$$

Finally, $\langle \mathbf{u}_k | \mathbf{u}_k \rangle = \sum_{n=0}^{N-1} \rho^{(k-k)n} = N$. □

The discrete Fourier transform

- According to the Fourier Basis Theorem, every $\mathbf{x} \in \mathbb{C}^N$ has the following **Fourier representation** (see the Exercises for more details)

$$\mathbf{x} = \frac{1}{N} \sum_{k=0}^{N-1} \langle \mathbf{x} | \mathbf{u}_k \rangle \mathbf{u}_k =: \frac{1}{N} \sum_{k=0}^{N-1} X(k) \mathbf{u}_k.$$

- $\hat{\mathbf{x}} \equiv \mathbf{X} := (X(0), X(1), \dots, X(N-1))^T \in \mathbb{C}^N$ denotes the vector of **Fourier coefficients**, where the k th coefficient is defined by

$$X(k) := \langle \mathbf{x} | \mathbf{u}_k \rangle = \sum_{n=0}^{N-1} x(n) \overline{\exp(2\pi i k n / N)} = \sum_{n=0}^{N-1} x(n) \exp(-2\pi i k n / N).$$

- By definition, the **discrete Fourier transform (DFT)** is the linear mapping $\mathbb{C}^N \rightarrow \mathbb{C}^N$ which maps $\mathbf{x}$ to $\mathbf{X}$.
- The DFT can be described by the $N \times N$ -matrix DFT_N given by

$$\text{DFT}_N(k, n) = \exp(-2\pi i k n / N) = \exp(-2\pi i / N)^{k \cdot n}.$$

DFT matrices

- Let $\omega := \exp(-2\pi i/N)$. Then $\text{DFT}_N(k, n) = \omega^{k \cdot n}$. Thus

$$\text{DFT}_N = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \dots & \omega^{(N-1)(N-1)} \end{pmatrix}.$$

- Note that for $\mathbf{x} \in \mathbb{C}^N$ we have $\boxed{\text{DFT}_N \mathbf{x} = \mathbf{X}}$.

- Here are some specific examples:

$$\text{DFT}_1 = (1), \quad \text{DFT}_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \text{DFT}_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}.$$

DFT: fundamental properties (1)

Theorem 2. *DFT matrices have the following properties:*

- (1) $\boxed{\text{DFT}_N \text{ is invertible.}}$ More precisely, with $\rho := \bar{\omega} = \exp(2\pi i/N)$ it holds

$$\text{DFT}_N^{-1} = \frac{1}{N} \cdot \overline{\text{DFT}_N}^\top = \frac{1}{N} \cdot \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \rho & \rho^2 & \dots & \rho^{N-1} \\ 1 & \rho^2 & \rho^4 & \dots & \rho^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \rho^{N-1} & \rho^{2(N-1)} & \dots & \rho^{(N-1)(N-1)} \end{pmatrix}.$$

- (2) $\boxed{\text{DFT}_N^2 = N \cdot P_N}$, where P_N denotes the $N \times N$ permutation matrix corresponding to the permutation $0 \mapsto 0$ and $k \mapsto N - k$, for $k \in [1 : N - 1]$:

$$\text{DFT}_N^2 = N \cdot \begin{pmatrix} 1 & & & & \\ & & & 1 & \\ & & 1 & & \\ & & & \ddots & \\ & 1 & & & \end{pmatrix}.$$

Proofs of DFT properties (1)

- (1): We have to show that $(\omega^{kn}) \cdot N^{-1} \cdot (\omega^{-n\ell})$ is the $N \times N$ identity matrix. But the entry at position (k, ℓ) of this product equals $N^{-1} \sum_{n=0}^{N-1} \omega^{(k-\ell)n} = \delta_{k,\ell}$.
- (2): For the entry at position (k, ℓ) in DFT_N^2 we obtain

$$\sum_{n=0}^{N-1} \omega^{(k+\ell)n} = N, \quad \text{if } k + \ell \equiv 0 \pmod{N},$$

otherwise the sum equals zero. But $k + \ell \equiv 0 \pmod{N}$ iff $k = \ell = 0$ or $k + \ell = N$, i.e., $\ell = N - k$. $\square$

Remark. As $\text{DFT}_N^2 = N \cdot P_N$ and P_N is self-inverse, i.e., $P_N = P_N^{-1}$, we obtain

$$\boxed{\text{DFT}_N^{-1} = \frac{1}{N} \cdot \text{DFT}_N \cdot P_N}.$$

Spectrum, magnitude spectrum, and power spectrum

We introduce fundamental definitions, closely related to the DFT of a signal.

Definition 3. Let $x \in \mathbb{C}^N$. Then we call

- $X = \hat{x} := \text{DFT}_N x$ the **spectrum** of x ,
- $|X| := (|X(0)|, |X(1)|, \dots, |X(N-1)|)^\top$ the **magnitude spectrum** of x , and
- $|X|^2 := (|X(0)|^2, |X(1)|^2, \dots, |X(N-1)|^2)^\top$ the **power spectrum** of x . $\diamond$

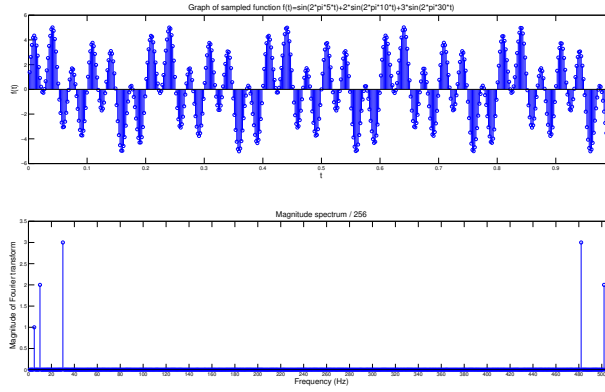
Example. Let $N = 512 = 2^9$. We consider

$$\boxed{f(t) := \sin(2\pi 5t) + 2 \cdot \sin(2\pi 10t) + 3 \cdot \sin(2\pi 30t)} \quad \text{for } t \in \{k/N \mid k \in [0 : N-1]\}.$$

- This is a weighted superposition of three sine waves of 5 Hz, 10 Hz, and 30 Hz, and corresponds to $\mathbf{f} = (\mathbf{f}_0, \dots, \mathbf{f}_{N-1})^\top \in \mathbb{C}^N$, where $\mathbf{f}_k := f(k/512)$.
- With $\sin(z) = (e^{iz} - e^{-iz})/(2i)$ and $\mathbf{u}_k(n) = \exp(2\pi i k n / N)$ we obtain

$$\boxed{\mathbf{f} = \frac{1}{2i}(\mathbf{u}_5 - \mathbf{u}_{N-5}) + \frac{2}{2i}(\mathbf{u}_{10} - \mathbf{u}_{N-10}) + \frac{3}{2i}(\mathbf{u}_{30} - \mathbf{u}_{N-30})}. \quad (1)$$

Example of a magnitude spectrum



- Combining $\mathbf{f} = \frac{1}{2i}(\mathbf{u}_5 - \mathbf{u}_{N-5}) + \frac{2}{2i}(\mathbf{u}_{10} - \mathbf{u}_{N-10}) + \frac{3}{2i}(\mathbf{u}_{30} - \mathbf{u}_{N-30})$ with $\langle \mathbf{u}_k | \mathbf{u}_\ell \rangle = N \cdot \delta_{k,\ell}$, we obtain $|\hat{\mathbf{f}}(k)| = w_k \cdot \frac{N}{2}$, where $w_5 = w_{N-5} = 1$, $w_{10} = w_{N-10} = 2$, $w_{30} = w_{N-30} = 3$, and $w_k = 0$ otherwise. Thus $\frac{2}{N}|\hat{\mathbf{f}}(k)| = |\hat{\mathbf{f}}(k)|/256 = w_k$, in accordance with the above stem plot.

DFT: fundamental properties (2)

Theorem 4. Let $\mathbf{X}$ and $\mathbf{Y}$ denote the spectrum of $x, y \in \mathbb{C}^N$, respectively.

- (1) $\langle \mathbf{X} | \mathbf{Y} \rangle = N \cdot \langle x | y \rangle$.
- (2) If y denotes the m -fold cyclic shift of x , i.e., $y(n) := x(n - m \bmod N)$, then

$$\mathbf{Y} = \Delta_{N,m} \cdot \mathbf{X}, \quad \text{where } \Delta_{N,m} = \text{diag}(\omega^{km})_{k \in [0:N-1]} \text{ and } \omega = \exp(-2\pi i/N).$$

Thus $Y(k) = \omega^{km} X(k)$ for all $k \in [0 : N - 1]$. In particular, $|\mathbf{Y}| = |\mathbf{X}|$, i.e., a cyclic shift does not change the magnitude spectrum.

- (3) $X(k) = \overline{X(N-k)}$, for all $k \in [0 : N - 1]$, if and only if x is **real-valued**. Thus for a real-valued signal x only half of its spectrum is needed. Furthermore, the magnitude spectrum of $x \in \mathbb{R}^N$ is symmetric: $|X(k)| = |X(N-k)|$.
- (4) If y is the **cyclic time reversal** of $x \in \mathbb{R}^N$, i.e., $y(n) = x(N - n \bmod N)$, then $Y(k) = \overline{X(k)}$, for all $k \in [0 : N - 1]$. In particular, time reversal of a real-valued signal does not change the magnitude spectrum: $|\mathbf{X}| = |\mathbf{Y}|$.

Proofs of DFT properties (2)

- (1): As $\text{DFT}_N = \text{DFT}_N^\top$ and $(AB)^\top = B^\top A^\top$, Theorem 2 (1) yields

$$\langle X|Y \rangle = \mathbf{X}^\top \cdot \overline{\mathbf{Y}} = (\text{DFT}_N \mathbf{x})^\top \cdot \overline{\text{DFT}_N \mathbf{y}} = \mathbf{x}^\top \cdot \text{DFT}_N \overline{\text{DFT}_N} \cdot \overline{\mathbf{y}} = N \langle \mathbf{x} | \mathbf{y} \rangle.$$

- (2):

$$\begin{aligned} Y(k) &= \sum_{n=0}^{N-1} \omega^{kn} y(n) = \sum_{n=0}^{N-1} \omega^{kn} x(n - m \bmod N) \\ &= \sum_{n=0}^{N-1} \omega^{k(n-m+m)} x(n - m \bmod N) \\ &= \omega^{km} \sum_{n=0}^{N-1} \omega^{k(n-m \bmod N)} x(n - m \bmod N) = \omega^{km} X(k). \end{aligned}$$

Proofs of DFT properties (2) continued

- (3): For all $k \in [0 : N - 1]$,

$$\sum_{n < N} x(n) \omega^{kn} = X(k) = \overline{\overline{X(N-k)}} = \overline{\sum_{n < N} x(n) \omega^{(N-k)n}} = \sum_{n < N} \overline{x(n)} \omega^{kn}$$

is equivalent to $\text{DFT}_N \mathbf{x} = \text{DFT}_N \overline{\mathbf{x}}$. As DFT_N is invertible this is equivalent to $\mathbf{x} = \overline{\mathbf{x}}$, i.e., $\mathbf{x} \in \mathbb{R}^N$.

- (4): As $\mathbf{x}$ is real and $\omega^{kN} = 1$, we obtain

$$\begin{aligned} Y(k) &= \sum_{n=0}^{N-1} \omega^{kn} y(n) = \sum_{n=0}^{N-1} \omega^{kn} x(N - n \bmod N) \\ &= \sum_{n=0}^{N-1} \omega^{-k(N-n)} x(N - n \bmod N) \\ &= \sum_{n=0}^{N-1} \omega^{-kn} x(n) = \overline{\sum_{n=0}^{N-1} \omega^{kn} x(n)} \quad (\mathbf{x} \text{ is real}) \\ &= \overline{X(k)}. \end{aligned}$$

□

The Wiener Khinchin theorem

Definition 5. Let $x \in \mathbb{C}^N$. Then $\text{ACF}[x] \in \mathbb{C}^N$, the **autocorrelation** of x , is defined at $m \in [0 : N - 1]$ by

$$\text{ACF}[x](m) = \sum_{n=0}^{N-1} x(n) \overline{x(n - m \bmod N)} = \langle x | T_m x \rangle,$$

where $T_m x$ denotes the m -fold cyclic shift of x : $T_m x(n) := x(n - m \bmod N)$.

Theorem 6. (Wiener Khinchin) Let $x \in \mathbb{C}^N$. Then the spectrum of its autocorrelation equals the power spectrum of x : $\widehat{\text{ACF}[x]} = |\mathbf{X}|^2$.

Proof. Theorem 4 (1) & (2) yield

$$\begin{aligned} \text{ACF}[x](m) &= \langle x | T_m x \rangle \stackrel{(1)}{=} N^{-1} \langle \mathbf{X} | \widehat{T_m x} \rangle \\ &\stackrel{(2)}{=} N^{-1} \langle \mathbf{X} | \Delta_{N,m} \mathbf{X} \rangle = N^{-1} \sum_{k=0}^{N-1} \omega^{-km} |X(k)|^2, \end{aligned}$$

i.e., $\widehat{\text{ACF}[x]} = \text{DFT}_N^{-1} |\mathbf{X}|^2$. Thus $\widehat{\widehat{\text{ACF}[x]}} = \text{DFT}_N \text{ACF}[x] = |\mathbf{X}|^2$. □

Fast computation of autocorrelations

- The computation of autocorrelations (or variants thereof) is an indispensable tool for finding repeated patterns in a signal.
- One obvious way to compute $\text{ACF}[x]$ for an input signal $x \in \mathbb{C}^N$ is to calculate the inner products $\langle x | T_m x \rangle = \sum_{n=0}^{N-1} x(n) \overline{x(n - m \bmod N)}$ for all $m \in [0 : N - 1]$ separately. This amounts to $O(N^2)$ arithmetic operations.
- The formula $\widehat{\text{ACF}[x]} = \text{DFT}_N^{-1} |\mathbf{X}|^2$ indicates an alternative way to compute $\text{ACF}[x]$.
- Below we will see that multiplying DFT_N or DFT_N^{-1} with a vector in $\mathbb{C}^N$ can be done with $O(N \log N)$ arithmetic operations.
- Hence the pretended detour algorithm

$$x \mapsto \text{DFT}_N x =: \mathbf{X} \mapsto |\mathbf{X}|^2 \mapsto \text{DFT}_N^{-1} |\mathbf{X}|^2 = \widehat{\text{ACF}[x]}$$

is much faster, as this detour can be done with $O(N \log N)$ arithmetic operations.

Fast Fourier Transforms

- The naive method for computing $\mathbb{C}^N \ni x \mapsto \mathbf{X} = \text{DFT}_N x$, i.e., computing the spectrum $\mathbf{X}$ of a length N signal x , amounts to order N^2 arithmetic operations.
- This is too slow for many real-time DSP-applications.
- Based on the fact that DFT matrices are highly structured, it has been proved that spectra of length N signals can be computed in $O(N \log N)$.

Theorem 7. $\mathbb{C}^N \ni x \mapsto \mathbf{X} = \text{DFT}_N x$ can be computed with at most

- (1) $1.5N \log N - N + 1$ arithmetic operations, if N is a power of two.
 - (2) $8N \log N$ arithmetic operations, for arbitrary N .
- (1) can be traced back to Gauß (1805, published posthumously 1866), but has been made popular by Cooley & Tukey (1965).
 - (2) has been shown by Baum, Clausen, Tietz, based on work of Bluestein and Rader, see Chapter 4 in the book Clausen, Baum: **Fast Fourier Transforms**, BI Wissenschaftsverlag, 1993.

DFT as multiple polynomial evaluation

Theorem 8. Let $x = (x_0, x_1, \dots, x_{N-1})^\top \in \mathbb{C}^N$ and $\mathbf{X} = \text{DFT}_N x$. Define the polynomial $p(z) = \sum_{n=0}^{N-1} x_n z^n$ of degree $< N$. Then with $\omega = \exp(-2\pi i/N)$

$$\mathbf{X} = (p(\omega^0), p(\omega^1), p(\omega^2), \dots, p(\omega^{N-1}))^\top,$$

i.e., computing $\mathbf{X}$ is the same as evaluating the polynomial $p(z)$ at all N th roots of unity.

Proof. By definition, for all $k \in [0 : N - 1]$,

$$X(k) = \sum_{n=0}^{N-1} \omega^{kn} x_n = \sum_{n=0}^{N-1} x_n (\omega^k)^n = p(\omega^k).$$

This proves our claim. □

- As we will see in a moment, the Cooley-Tukey FFT is based on this multiple polynomial evaluation idea.

FFT: the basic idea

- We keep the above notation and suppose that $N = 2M$ is even. We split $p(z) = \sum_{n=0}^{N-1} x_n z^n$ into its even and odd parts:

$$\begin{aligned} p(z) &= \left(\sum_{m=0}^{M-1} x_{2m} z^{2m} \right) + z \cdot \left(\sum_{m=0}^{M-1} x_{2m+1} z^{2m} \right) \\ &= \left(\sum_{m=0}^{M-1} x_{2m} (z^2)^m \right) + z \cdot \left(\sum_{m=0}^{M-1} x_{2m+1} (z^2)^m \right). \end{aligned}$$

If we define $p_0(y) = \sum_{m=0}^{M-1} x_{2m} y^m$ and $p_1(y) = \sum_{m=0}^{M-1} x_{2m+1} y^m$, then

$$\boxed{p(z) = p_0(z^2) + z \cdot p_1(z^2)} \quad \text{and} \quad \boxed{p(-z) = p_0(z^2) - z \cdot p_1(z^2)}.$$

- Now $\mu := \omega^2 = \exp(-2\pi i/M)$ is a primitive M th root of unity. Furthermore, for $k \in [0 : M-1]$,

$$\boxed{\mu^k = \omega^{2k} = (\omega^k)^2 = (-\omega^k)^2 = (\omega^{M+k})^2}.$$
- Combining the framed formulas yields the Cooley-Tukey FFT.

The decimation-in-time Cooley-Tukey FFT

Theorem 9. Let $N = 2M$ and $\mathbf{x} \in \mathbb{C}^N$. If $\mathbf{x}^0 = (x_0, x_2, \dots, x_{N-2})^\top \in \mathbb{C}^M$ and $\mathbf{x}^1 = (x_1, x_3, \dots, x_{N-1})^\top \in \mathbb{C}^M$ denote the even and odd parts of $\mathbf{x}$ with corresponding spectra $\mathbf{X}^0$ and $\mathbf{X}^1$, and if $\Delta_M := \text{diag}(1, \omega^1, \dots, \omega^{M-1})$ denotes the diagonal matrix of twiddle factors (resulting from multiplications by z), then

$$\boxed{(X(0), \dots, X(M-1))^\top = \mathbf{X}^0 + \Delta_M \cdot \mathbf{X}^1} \quad (\text{upper half of } \mathbf{X})$$

$$\boxed{(X(M), \dots, X(N-1))^\top = \mathbf{X}^0 - \Delta_M \cdot \mathbf{X}^1} \quad (\text{lower half of } \mathbf{X}).$$

Proof. Keeping the above notation, we obtain for $k \in [0 : M-1]$

$$X(k) = p(\omega^k) = p_0(\mu^k) + \omega^k \cdot p_1(\mu^k) = X^0(k) + \omega^k \cdot X^1(k),$$

and

$$\begin{aligned} X(M+k) &= p(\omega^{M+k}) = p(-\omega^k) = p_0(\mu^k) - \omega^k \cdot p_1(\mu^k) \\ &= X^0(k) - \omega^k \cdot X^1(k), \end{aligned}$$

which proves our claims. □

Cooley-Tukey FFT: complexity analysis

Theorem 10. *If N is a power of two, the decimation-in-time Cooley-Tukey FFT computes the spectrum of $x \in \mathbb{C}^N$ with at most $1.5N \log N - N + 1$ arithmetic operations.*

Proof. Let $N = 2M = 2^k$. If $T(N)$ denotes the number of arithmetic operations of the Cooley-Tukey FFT applied to an arbitrary vector in $\mathbb{C}^N$, then

$$T(N) \leq 2 \cdot T(M) + (M - 1) + N.$$

- the term $2 \cdot T(M)$ is for computing $\mathbf{X}^0$ and $\mathbf{X}^1$,
- $(M - 1)$ is the cost for multiplying $\mathbf{X}^1$ with $\Delta_M = \text{diag}(1, \omega^1, \dots, \omega^{M-1})$,
- finally one has to perform N additions/subtractions.

As $N = 2^k$, we can iterate this procedure. Starting with $T(1) = 0$, an easy induction shows that $T(N) \leq 1.5N \log N - N + 1$. □

Remark. FFTs for more general lengths will be discussed in the Exercises.

Audio denoising (1)

- Audio signals are often contaminated with noise (channel, background, buzzing, etc.).
- Audio denoising aims at attenuating the noise while retaining the underlying signals.
- The aim of the next slides is to indicate – within a simplistic setting – the use of DFT/FFT in combination with thresholding for noise reduction.
- To this end we start with a pure sine wave of 4 Hz.

$$\mathbf{a} = (\sin(2\pi 4k/N))_{k \in [0: N-1]}, \quad \text{where } N = 512.$$

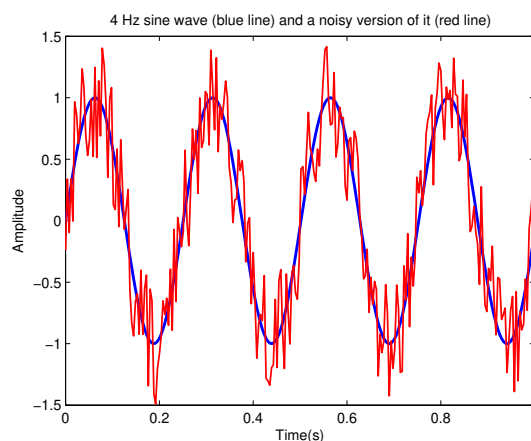
- This source signal is superimposed by additive noise. Such a noise signal $\mathbf{r}$ can be defined in MATLAB by

$$\mathbf{r} = \text{rand}(N, 1) - 0.5.$$

Thus $\mathbf{r}$ is an N -tuple of pseudo random numbers, drawn uniformly and independently from the real interval $(-0.5, +0.5)$. Hence we receive the noisy signal $\mathbf{b} := \mathbf{a} + \mathbf{r}$.

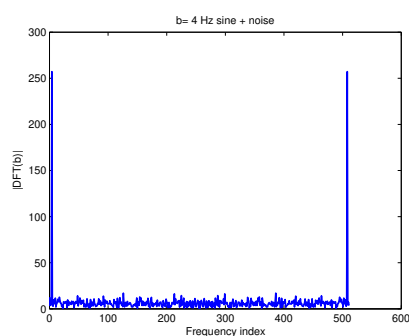
Audio denoising (2)

- The following figure shows the underlying signal a (in blue) as well as its noisy version b (in red).



Audio denoising (3)

- Typically, the magnitude spectrum $|R|$ of such a noise signal r is flat. On the other hand, the energy of our source signal a is concentrated on a few frequencies (4 Hz and 508 Hz, in our case).
- So in the magnitude spectrum $|B|$ of the noisy signal $b = a + r$ we expect two large values and only small values everywhere else.
- The following figure confirms our argumentation.



Audio denoising (4)

- By **hard thresholding** we set any coefficient in $\mathbf{B}$ whose absolute value is less than or equal to a threshold θ to zero. All other values of $\mathbf{B}$ remain unchanged. The resulting spectral vector is denoted by $\mathbf{B}_\theta$.
- In our case, for suitably chosen θ (choice of θ is a delicate issue!), we obtain via inverse Fourier transform:

$$\text{DFT}_N^{-1} \mathbf{B}_\theta = \mathbf{a} + \mathbf{d},$$

where $\mathbf{d}$ denotes a tiny restoration error. Details will be discussed in the exercises.

- **Warning:** The fact that we can reconstruct $\mathbf{a}$ – up to a tiny perturbation – is due to the very special input signal!
- There are better – but more involved – denoising methods.

DFT: fundamental properties (3)

For signals $\mathbf{x}, \mathbf{y}$ in $\mathbb{C}^N$ we define two kinds of multiplication,

- **pointwise multiplication:** $\mathbf{x} \odot \mathbf{y}$, defined for $k \in [0 : N - 1]$ by

$$(\mathbf{x} \odot \mathbf{y})(k) := x(k) \cdot y(k).$$

- **cyclic convolution:** $\mathbf{x} * \mathbf{y}$, defined for $k \in [0 : N - 1]$ by

$$(\mathbf{x} * \mathbf{y})(k) := \sum_{n=0}^{N-1} x(n) \cdot y(k - n \bmod N).$$

Both kinds of multiplication are of great importance in DSP:

- **pointwise multiplication** is the basis of **windowing**, whereas
- **cyclic convolution** is closely related to **filtering**.

We clarify the impact of these multiplications in the spectral domain.

Theorem 11. For $\mathbf{x}, \mathbf{y}$ in $\mathbb{C}^N$ the following holds.

- (a) $\text{DFT}_N(\mathbf{x} * \mathbf{y}) = \text{DFT}_N \mathbf{x} \odot \text{DFT}_N \mathbf{y}$.
- (b) $\text{DFT}_N(\mathbf{x} \odot \mathbf{y}) = \frac{1}{N} \cdot \text{DFT}_N \mathbf{x} * \text{DFT}_N \mathbf{y}$.

Pointwise multiplication and cyclic convolution

- Before proving these results, we comment on it.
- $\widehat{x * y} = \widehat{x} \odot \widehat{y}$ is called the **Convolution Theorem** (for signals of length N). It says that cyclic convolution in the time domain corresponds to pointwise multiplication in the spectral domain. This fact is the basis of **fast polynomial multiplication**, which is among the fundamental computational DSP problems. The Convolution Theorem signifies that the spectrum of an input signal x can be manipulated pointwise by a cyclic convolution with a suitable y . This kind of frequency manipulations, called **(convolutional) filtering**, will be further discussed in subsequent chapters.
- $\widehat{x \odot y} = N^{-1} \cdot \widehat{x} * \widehat{y}$ says that pointwise multiplication in the time domain corresponds to cyclic convolution in the spectral domain. E.g., multiplying an input signal x pointwise with a box function $y = e_a + e_{a+1} + \dots + e_b$, $0 \leq a < b < N$, results in a **windowed version** $x \odot y$ of x . Thus the spectrum of a windowed version of a signal x is a scalar multiple of the convolution of the original spectrum $\widehat{x}$ with the spectrum $\widehat{y}$ of the window. After proving the above theorem, the benefit of such a windowing technique will be our next topic.

Proofs of DFT properties (3)

(a) For $k \in [0 : N - 1]$ we obtain ($\sum_{n < N}$ means $\sum_{n=0}^{N-1}$)

$$\begin{aligned}
 \text{DFT}_N(x * y)(k) &= \sum_{n < N} \omega^{kn} (x * y)(n) \\
 &= \sum_{n < N} \omega^{kn} \sum_{m < N} x(m) y(n - m \bmod N) \\
 &= \sum_{m < N} \sum_{n < N} \omega^{k(n-m+m)} x(m) y(n - m \bmod N) \\
 &= \sum_{m < N} \omega^{km} x(m) \sum_{n < N} \omega^{k(n-m \bmod N)} y(n - m \bmod N) \\
 &= \sum_{m < N} \omega^{km} x(m) \sum_{n < N} \omega^{kn} y(n) \\
 &= (\text{DFT}_N x)(k) \cdot (\text{DFT}_N y)(k) \\
 &= (\text{DFT}_N x \odot \text{DFT}_N y)(k).
 \end{aligned}$$

Thus $\text{DFT}_N(x * y) = \text{DFT}_N x \odot \text{DFT}_N y$, which proves (a).

Proofs of DFT properties (3) continued

(b) For $k \in [0 : N - 1]$ we obtain

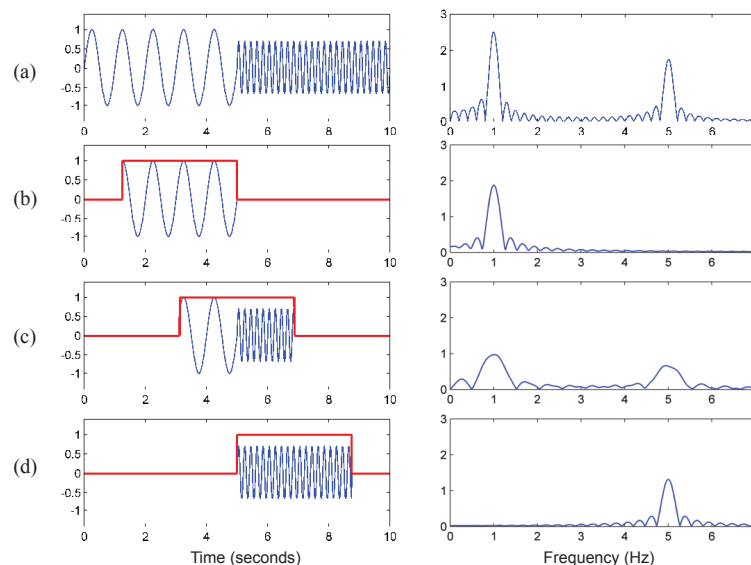
$$\begin{aligned}
 \frac{1}{N}(\text{DFT}_N \mathbf{x} * \text{DFT}_N \mathbf{y})(k) &= \frac{1}{N} \sum_{n < N} (\text{DFT}_N \mathbf{x})(n) (\text{DFT}_N \mathbf{y})(k - n \bmod N) \\
 &= \frac{1}{N} \sum_{n < N} \sum_{p < N} \omega^{np} x(p) \sum_{q < N} \omega^{(k-n \bmod N)q} y(q) \\
 &= \frac{1}{N} \sum_{n < N} \sum_{p < N} \omega^{np} x(p) \sum_{q < N} \omega^{(k-n)q} y(q) \\
 &= \sum_{p < N} \sum_{q < N} x(p) y(q) \omega^{kq} \cdot \underbrace{\frac{1}{N} \sum_{n < N} \omega^{(p-q)n}}_{=\delta_{p,q}} \\
 &= \sum_{p < N} x(p) y(p) \omega^{kp} = \sum_{p < N} (\mathbf{x} \odot \mathbf{y})(p) \omega^{kp} \\
 &= (\text{DFT}_N(\mathbf{x} \odot \mathbf{y}))(k).
 \end{aligned}$$

Thus $\text{DFT}_N(\mathbf{x} \odot \mathbf{y}) = N^{-1} \cdot \text{DFT}_N \mathbf{x} * \text{DFT}_N \mathbf{y}$, which proves (b). □

STFT: a local-global compromise

- Up to now we have looked at discrete signals from two extreme points of view:
 - ◆ the **extremely local viewpoint** based on unit vectors $e_0, \dots, e_{N-1}$ in $\mathbb{C}^N$, and
 - ◆ the **extremely global viewpoint** based on the Fourier basis $\mathbf{u}_0/\sqrt{N}, \dots, \mathbf{u}_{N-1}/\sqrt{N}$ in $\mathbb{C}^N$.
- One major drawback of the **local viewpoint** is the fact that it **hides frequency information**.
- One major drawback of the **global viewpoint** is the fact that it **hides time information**.
- The **short-time Fourier transform** (STFT) combines these extreme viewpoints by taking **global views** at **small sections** of the signal and receives **simultaneously time and frequency informations**.
- Small sections are obtained by multiplying the signal pointwise with time-shifted versions of a **window function**, which is a function that is non-zero for only a short period of time.

STFT: an example



- For the moment, we will consider analog (continuous-time) signals.

STFT: comment on the example

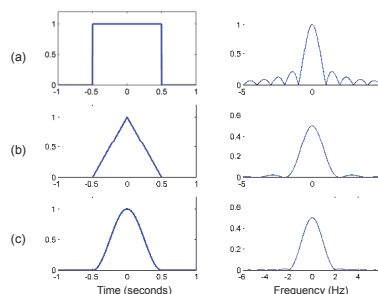
- The above figure is about two subsequent sinusoids of 1 Hz and 5 Hz.
- (a) shows the original signal and its magnitude spectrum. This spectrum indicates that sinusoids of 1 Hz and 5 Hz are present in the original signal. However, the information when these frequencies occur is hidden.
- As window function we use a **rectangular function** (also called **box function**) which takes 4 seconds. The shifted versions of the box functions are shown in red.
- Multiplying the original signal pointwise with the shifted windows yields **windowed signals** centered at time $t = 3$, $t = 5$, and $t = 7$, see the first column of (b) – (d).
- The second column of (b) – (d) shows the corresponding magnitude spectra.
- In contrast to (a), the magnitude spectrum in (b) (resp. (d)) shows that a 1 Hz (resp. 5 Hz) sinusoid is dominant in the immediate vicinity of $t = 3$ (resp. $t = 7$). (c) shows that both a 1 Hz and a 5 Hz sinusoid are present near $t = 5$.

Window functions

- To soften artifacts caused by the discontinuities of the box functions, one often uses windows that are non-negative within the desired section and continuously fall to zero towards the section's boundaries. One such example is the **triangular window**, see below. A window often used in DSP is the **Hann window**, defined by

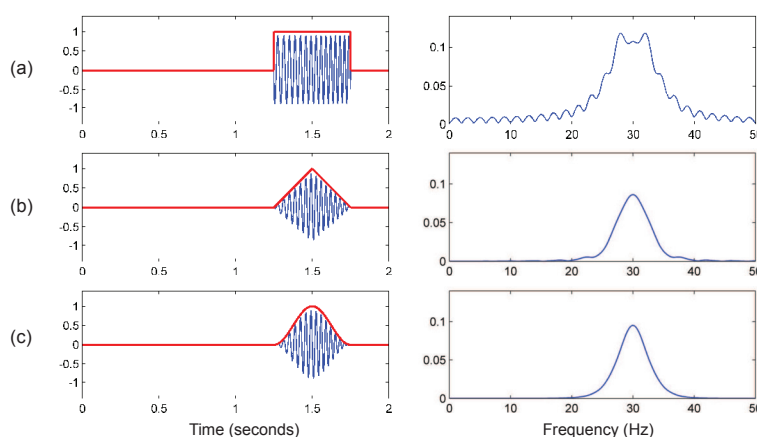
$$g(u) := \frac{1}{2} \cdot (1 + \cos(\pi u)), \text{ for } |u| \leq 0.5 \quad \text{and} \quad g(u) := 0 \quad \text{otherwise.}$$

- The following figure shows three windows ((a) box, (b) triangular, (c) Hann) and their magnitude spectra.



STFT: the impact of window functions

- The following figure shows a windowed chirp signal and its magnitude spectrum using different window functions: (a) box, (b) triangular, (c) Hann window.
- The use of a Hann window shows a clear reduction of artifacts.



Discrete STFT

- After this short trip into the analog world, we come back to the discrete setting.
- Let $x : \mathbb{Z} \rightarrow \mathbb{C}$ denote a discrete time signal, whose **support** $\text{supp}(x) := \{n \in \mathbb{Z} \mid x(n) \neq 0\}$ is finite.
- Let $w = (w(0), w(1), \dots, w(N-1))^\top$ denote a **discrete window function**.
- Furthermore, let $H \in [1 : N-1]$ denote a fixed **hop size**.
- For each $m \in \mathbb{Z}$, we define
 - ◆ the m th **windowed frame** $x_m = (x_m(0), \dots, x_m(N-1))^\top \in \mathbb{C}^N$ by

$$x_m(n) := x(n + Hm) \overline{w(n)}, \quad \text{for } n \in [0 : N-1]$$
 - ◆ and its spectrum $X_m = (X_m(0), \dots, X_m(N-1))^\top$ by

$$X_m := \text{DFT}_N \cdot x_m.$$
- If both x and w are real-valued, the upper half of the frequency coefficients is redundant.

From CT- via DT- to finite length signals (1)

- So far, we have considered mainly signals of finite length.
- Typically, one starts with a continuous-time signal $f : \mathbb{R} \rightarrow \mathbb{C}$.
- Under certain integrability conditions ($f \in L^2(\mathbb{R})$), f possesses a Fourier transform

$$\hat{f}(\omega) = \int_{\mathbb{R}} f(t) \cdot e^{-2\pi i \omega t} dt, \quad \text{for } \omega \in \mathbb{R}.$$

- If f is **Ω -bandlimited**, i.e., $\hat{f}(\omega) = 0$, for all $|\omega| > \Omega$, and the **sampling period** T satisfies $0 < T \leq 1/(2\Omega)$, then the T -sampled discrete-time signal $x : \mathbb{Z} \rightarrow \mathbb{C}$ defined by $x(n) := f(T \cdot n)$ allows to reconstruct f with the help of the **sinc-function**: $\text{sinc}(t) := \sin(\pi t)/(\pi t)$.

Sampling Theorem:
$$f(t) = \sum_{n \in \mathbb{Z}} f(nT) \cdot \text{sinc}\left(\frac{t}{T} - n\right).$$

- Under mild conditions ($x \in \ell^2(\mathbb{Z})$), also x possesses a Fourier transform

$$\hat{x}(\omega) = \sum_{n \in \mathbb{Z}} x(n) \cdot e^{-2\pi i \omega n}, \quad \text{for } \omega \in [0, 1).$$

From CT- via DT- to finite length signals (2)

- Fixing a number N of samples, one can compute $\mathbf{X} = \text{DFT}_N \cdot \mathbf{x}$ for $\mathbf{x} = (x(0), x(1), \dots, x(N-1))^T$.
- **Question:** What is the meaning of the Fourier coefficients $\mathbf{X} = (X(0), X(1), \dots, X(N-1))^T$ in relation to the original signal f ?
- Via Riemann sum approximation and the substitution rule we obtain

$$\begin{aligned}\hat{x}(\omega) &= \sum_{n \in \mathbb{Z}} x(n) \cdot e^{-2\pi i \omega n} = \sum_{n \in \mathbb{Z}} f(nT) \cdot e^{-2\pi i \omega n} \\ &\approx \int_{\mathbb{R}} f(t) e^{-2\pi i \omega t} dt = \frac{1}{T} \int_{\mathbb{R}} f(t) e^{-2\pi i \omega t/T} dt = \frac{1}{T} \hat{f}\left(\frac{\omega}{T}\right).\end{aligned}$$

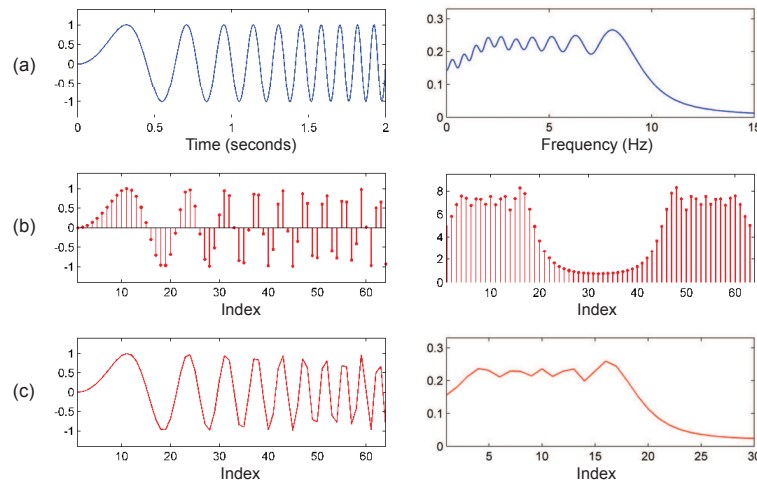
- On the other hand, $1/N$ -sampling of $\hat{x}$ yields for $k \in [0 : N-1]$ if $\sum_{n \notin [0:N-1]} |x(n)| \approx 0$:

$$\hat{x}(k/N) \approx \sum_{n=0}^{N-1} x(n) e^{-2\pi i kn/N} = X(k).$$

- Combining both approximations this results in

$$X(k) \approx \hat{x}\left(\frac{k}{N}\right) \approx \frac{1}{T} \cdot \hat{f}\left(\frac{k}{N} \cdot \frac{1}{T}\right). \quad (2)$$

DFT-approximation of CT-Fourier transform (1)



DFT-approximation of CT-Fourier transform (2)

- (a) shows plots of the analog chirp signal $f := ([0, 2] \ni t \mapsto \sin(2\pi t^2))$ and its continuous Fourier transform (shown for frequencies $\omega \in [0, 15]$).
- Next we sample f with sample period $T = 1/32$, i.e., with **sampling rate** $F_s = 1/T = 32$ Hz and obtain a finite-length signal x of length $N = 64$. Applying DFT_N results in a complex-valued vector $\mathbf{X} = \text{DFT}_N \cdot x$. Both x and its magnitude spectrum $|\mathbf{X}|$ are shown in (b).
- By Eq. (2) we obtain $X(k)/32 \approx \hat{f}(k/2)$. For example, the index $k = 30$ corresponds to frequency $\omega = 15$.
- In our example, the resulting **frequency resolution** $1/(NT)$ is 0.5 Hz.
- (c) shows the interpolation of the sampled signal and of the DFT coefficients.

Spectrogram representation

- If the STFT of a DT-signal $x : \mathbb{Z} \rightarrow \mathbb{R}$ (obtained by T -sampling) uses a window $w \in \mathbb{R}^N$ of length N and a hop size $H \in [1 : N - 1]$, then

$$\mathbf{X}(m, k) = \sum_{n < N} x(n + mH)w(n)e^{-2\pi i k n / N}$$

is the discrete STFT of x for the **frame index** m and the **frequency index** $k \in [0 : K]$, where $K = N/2$ (to avoid redundancies because both x and w are real-valued!). Note that k corresponds to the frequency $k/(TN)$.

- The corresponding **(power) spectrogram** is defined by

$$(m, k) \mapsto |\mathbf{X}(m, k)|^2.$$

- When generating an image of the spectrogram, the horizontal axis represents time, the vertical axis is frequency, and the dimension indicating spectrogram values is represented by the intensity of color in the image.

Logarithmic scales

- To emphasize musical and tonal relationships, the frequency axis of a spectrogram representation is often plotted in a logarithmic fashion, yielding a **log-frequency representation**. Such a representation also accounts for the fact that human perception of pitch is logarithmic in nature.
- Finally, in the case of audio signals, the amplitude values are also often visualized using a logarithmic scale, e.g., the **decibel scale**. Recall that the decibel function $\text{dB}: \mathbb{R}_{>0} \rightarrow \mathbb{R}$ is defined by

$$\text{dB}(r) := 20 \log_{10}(r).$$

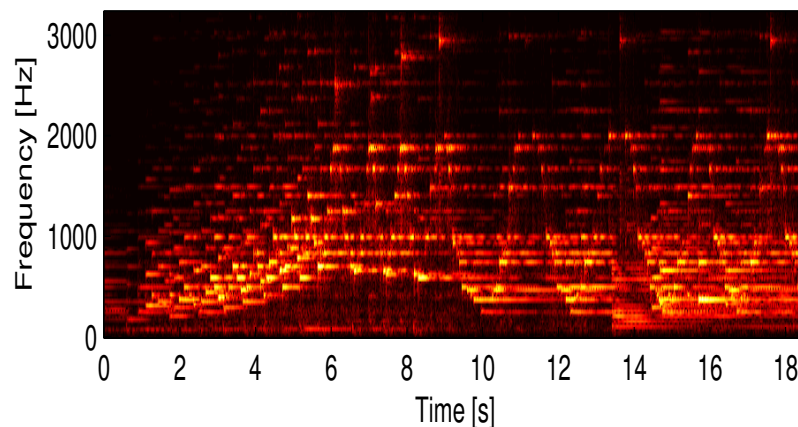
Note that the decibel function for $r \rightarrow 0$ is strictly decreasing; more precisely, $\text{dB}(1) = 0$ and $\lim_{r \rightarrow 0} \text{dB}(r) = -\infty$. Furthermore,

$$\text{dB}(r \cdot s) = \text{dB}(r) + \text{dB}(s), \quad \text{dB}(1000) = 60, \quad \text{dB}(2r) \approx 6 + \text{dB}(r).$$

With the dB-scale, small intensity values of perceptual relevance become visible in the image.

Spectrogram of a music signal

The following figure shows the spectrogram (with a Hann window) of a section of Claude Debussy's *Reflets dans l'eau* (interpreter: A. Brachet)



Spectrogram of a birdsong

The following figure shows the spectrogram of a strophe sung by a male chaffinch.

